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The number of legislators who voted for the bill was 8 more than the total number who were absent or abstained.

Arithmetic Equations

Since there are 96 members and 3 + 25 = 28 of the members did not vote, it follows that 96 – 28 = 68 of the members voted. We are to determine the number of members who voted FOR the bill.

1. We are given that $\frac{1}{3}\left(96\right) = 32$ members voted AGAINST the bill. Therefore, 68 – 32 = 36 members voted FOR the bill; SUFFICIENT.
2. We are given that 8 + (25 + 3) = 36 members voted FOR the bill; SUFFICIENT.

The correct answer is D; each statement alone is sufficient.

•

What is the average (arithmetic mean) of the first 90 measurements in a certain list of 92 measurements?

1. The average of the 92 measurements is 7.0 centimeters.
2. The average of the last 2 measurements is 7.2 centimeters.

Arithmetic Statistics

1. Given that the average of all 92 measurements is 7, it follows that the sum of all 92 measurements is 92 × 7 = 644. However, the sum of the first 90 measurements could be any number and hence we cannot determine the average of the first 90 measurements. For example, it is possible that the average of the first 90 measurements is 1 (sum of first 90 measurements is 90 and each of the last 2 measurements is 277), and it is possible that the average of the first 90 measurements is 7 (sum of first 90 measurements is 630 and each of the last 2 measurements is 7); NOT sufficient.
2. Given that the average of the last 2 measurements is 7.2, it follows that the sum of the last 2 measurements is 2 × 7.2 = 14.4. However, the sum of the first 90 measurements, and hence the average of the first 90 measurements, could be any number, and hence we cannot determine the average of the first 90 measurements; NOT sufficient.

Taking (1) and (2) together, the sum of the first 90 measurements is the sum of all 92 measurements minus the sum of the last 2 measurements, or 644 – 14.4 = 629.6. Therefore, the average of the first 90 measurements is 629.6 ÷ 90.

The correct answer is C; both statements together are sufficient.

•

How many of the employees of a certain company use both a tablet and an iPhone?

1. 60 of the employees use a tablet.
2. 45 of the employees use an iPhone.

Arithmetic Statistics

1. Given that 60 of the employees use a tablet, the number of employees who use both a tablet and an iPhone cannot be determined because no information about iPhone use by employees is provided; NOT sufficient.
2. Given that 45 of the employees use an iPhone, the number of employees who use both a tablet and an iPhone cannot be determined because no information about tablet use by employees is provided; NOT sufficient.

Taking (1) and (2) together, it is still not possible to determine the number of employees who use both a tablet and an iPhone. Indeed, the number of employees who use both could be any integer from 0 to 45, inclusive.

The correct answer is E; both statements together are still not sufficient.

•

What is the standard deviation of the weights of the 30 samples in a certain experiment?

1. The total weight of the 30 samples is 360 grams.
2. Each of the 30 samples weighs 12 grams.

Arithmetic Statistics

1. Given that the total weight of the 30 samples is 360 grams, the standard deviation of the weights cannot be

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1. Given that the total weight of the 30 samples is 360 grams, the standard deviation of the weights cannot be determined. For example, if each of the 30 samples weighed 12 grams (this gives a total weight of 360 grams), then the average weight would be $\frac{360}{30} = 12$ grams, each of the squared distances of the weights from the average weight would be 0, and the average of these squared distances would be 0. Therefore, the standard deviation of the weights—the square root of the average of these squared distances—would be $\sqrt{0} = 0$. On the other hand, if there were 15 samples that each weighed 11 grams and 15 samples that each weighed 13 grams (this gives a total weight of 360 grams), then the average weight would be $\frac{360}{30} = 12$ grams, each of the squared distances of the weights from the average weight would be 1, and the average of these squared distances would be 1. Therefore, the standard deviation of the weights—the square root of the average of these squared distances—would be $\sqrt{1} = 1$; NOT sufficient.
2. Given that each of the 30 samples weighs 12 grams, the first example above shows the standard deviation of the weights is 0; SUFFICIENT.

The correct answer is B; statement 2 alone is sufficient.

•

The people in a line waiting to buy tickets to a show are standing one behind the other. Adam and Beth are among the people in the line, and Beth is standing behind Adam with a number of people between them. If the number of people in front of Adam plus the number of people behind Beth is 18, how many people in the line are behind Beth?

1. There are a total of 32 people in the line.
2. 23 people in the line are behind Adam.

Arithmetic Order

Beth is standing in line behind Adam with a number of people between them. Let x be the number of people ahead of Adam, let y be the number of people between Adam and Beth, and let z be the number of people behind Beth. It is given that $x + z = 18$. Determine z.

1. This indicates that there are 32 people in the line. Two of these people are Adam and Beth, so there are 30 other people in line besides Adam and Beth. Therefore, $x + y + z = 30$ and $x + z = 18$. From this, $y = 12$, but z cannot be determined uniquely. For example, if $x = 5$, then $z = 13$, but if $x = 10$, then $z = 8$; NOT sufficient.
2. This indicates that $y + 1 + z = 23$ because the people behind Adam consist of the people between Adam and Beth, Beth herself, and the people behind Beth. If, for example, $y = 4$, then $z = 18$, but if $y = 9$, then $z = 13$; NOT sufficient.

Taking (1) and (2) together, $y = 12$ from (1) and $y + z = 22$ from (2). Therefore, $z = 10$, and there are 10 people in line behind Beth.

The correct answer is C; both statements together are sufficient.

•

In a certain group of 50 people, how many are doctors who have a law degree?

1. In the group, 36 people are doctors.
2. In the group, 18 people have a law degree.

Arithmetic Sets

1. Given that there are 36 people who are doctors, there is no information about how many of the 36 people have a law degree, and therefore it cannot be determined how many of the 36 people are doctors with a law degree; NOT sufficient.
2. Given that there are 18 people who have a law degree, there is no information about how many of the 18 people are doctors, and therefore it cannot be determined how many of the 18 people are doctors with a law degree; NOT sufficient.

Taking (1) and (2) together, if n is the number of doctors with a law degree, the contingency table below shows that more than one value of n is possible by assigning appropriate values in the column labeled "not a doctor."

	doctor	not a doctor	total

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	doctor	not a doctor	total
law degree	n		18
no law degree	$36-n$		32
total	36	14	50

For example, the first table below gives $n=10$ and the second table below gives $n=12$.

	doctor	not a doctor	total
law degree	n	8	18
no law degree	$36-n$	6	32
total	36	14	50

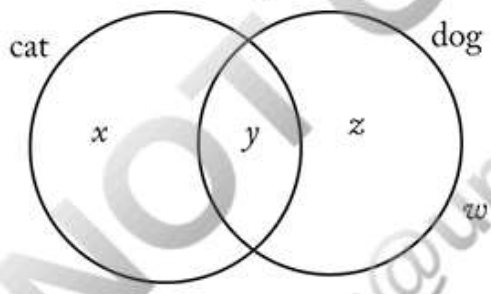
	doctor	not a doctor	total
law degree	n	6	18
no law degree	$36-n$	8	32
total	36	14	50

The correct answer is E; both statements together are still not sufficient.

•

Of a group of 50 households, how many have at least one cat or at least one dog, but not both?

1. The number of households that have at least one cat and at least one dog is 4.
2. The number of households that have no cats and no dogs is 14.



Arithmetic Sets

Using the labels on the Venn diagram above, $w+x+y+z=50$. Determine the value of $x+z$.

1. This indicates that $y=4$, so $w+x+z=46$, but the value of $x+z$ cannot be uniquely determined; NOT sufficient.
2. This indicates that $w=14$, so $x+y+z=36$, but the value of $x+z$ cannot be uniquely determined; NOT sufficient.

Taking (1) and (2) together, $14+x+4+z=50$, so $x+z=50-14-4=32$.

The correct answer is C; both statements together are sufficient.

•

Robin invested a total of $12,000 in two investments, X and Y, so that the investments earned the same amount of simple annual interest. How many dollars did Robin invest in Investment Y?

1. Investment X paid 3 percent simple annual interest, and Investment Y paid 6 percent simple annual interest.
2. Robin invested more than $1,000 in Investment X.

Algebra Applied Problems

For Investments X and Y, let r_X and r_Y be the annual percentage interest rates, respectively, and let x and y be the investment amounts, respectively and in dollars. Then $x+y=12{,}000$ and $r_X \cdot x = r_Y \cdot y$. What is the value of y?

1. Given that $r_X=3$ and $r_Y=6$, then from $r_X \cdot x = r_Y \cdot y$, it follows that $3x=6y$, or $x=2y$. Therefore,

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$x + y = 12{,}000$ becomes $2y + y = 12{,}000$, or $3y = 12{,}000$, or $y = 4{,}000$; SUFFICIENT.

2. Given that $x > 1{,}000$, it is not possible to determine the value of y. For example, $y = 4{,}000$ is possible (choose $r_X = 3$, $r_Y = 6$, and $x = 8{,}000$), and $y = 6{,}000$ is possible (choose $r_X = 3$, $r_Y = 3$, and $x = 6{,}000$); NOT sufficient.

The correct answer is A; statement 1 alone is sufficient.

•

In a real estate office that employs n salespeople, f of them are females and x of the females are new employees. What is the value of n?

1. If an employee were randomly selected from the n employees, the probability of selecting a female would be $\frac{2}{3}$.
2. If an employee were randomly selected from the f female employees, the probability of selecting a new employee would be $\frac{1}{2}$.

Arithmetic Probability

1. Given that $\frac{f}{n} = \frac{2}{3}$, it is possible that $n = 3$ (choose $f = 2$ and $x = 1$), and it is possible that $n = 6$ (choose $f = 4$ and $x = 2$); NOT sufficient.
2. Given that $\frac{x}{f} = \frac{1}{2}$, the value of n cannot be determined because $\frac{x}{f} = \frac{1}{2}$ is true for each of the two choices of values of n, f, and x that were used in (1); NOT sufficient.

Taking (1) and (2) together is still not sufficient because both (1) and (2) hold for each of the two choices of values of n, f, and x that were used in (1).

The correct answer is E; both statements together are still not sufficient.

•

Is the total price of the items in Andrea's shopping cart less than $25.00?

1. There are 5 items in Andrea's shopping cart.
2. The price of the most expensive item in Andrea's shopping cart is $4.99.

Arithmetic Estimation

1. Given that there are 5 items in the cart, the total price of the items in the cart could be less than $25 (for example, if the price of each item was $1), and the total price of the items in the cart could be greater than $25 (for example, if the price of each item was $6); NOT sufficient.
2. Given that the price of the most expensive item in the cart was $4.99, the total price of the items in the cart could be less than $25 (for example, if the cart contained only 2 items: 1 item with price $4.99 and 1 item with price $4), and the total price of the items in the cart could be greater than $25 (for example, if the cart contained 1 item with price $4.99 and 6 items with price $4); NOT sufficient.

Taking (1) and (2) together, the total price of the 5 items in the cart is at most 5($4.99) = $24.95, which is less than $25.

The correct answer is C; both statements together are sufficient.

•

A certain plumber charges $92 for each job completed in 4 hours or less and $23 per hour for each job completed in more than 4 hours. If it took the plumber a total of 7 hours to complete two separate jobs, what was the total amount charged by the plumber for the two jobs?

1. The plumber charged $92 for one of the two jobs.
2. The plumber charged $138 for one of the two jobs.

Arithmetic Applied Problems

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Find the total amount charged for two jobs lasting a total of 7 hours if the plumber charges \$92 for a job lasting 4 hours or less and \$23 per hour for a job lasting more than 4 hours.

1. This indicates that one of the two jobs lasted 4 hours or less. If that job lasted 2 hours, then the other job lasted 5 hours, and the charge for that job would be (\$23)(5) = \$115, making the total for the two jobs \$92 + \$115 = \$207. However, if the jobs lasted 1 hour and 6 hours, respectively, then the total charge for the two jobs would be \$92 + (6)(\$23) = \$92 + \$138 = \$230; NOT sufficient.
2. This indicates that one job lasted $\frac{\$138}{\$23} = 6$ hours. Therefore, the other job lasted for 1 hour and the charge for that job was \$92, making the total charge for the two jobs \$92 + \$138 = \$230; SUFFICIENT.

The correct answer is B; statement 2 alone is sufficient.

•

If x and y are positive numbers, is $\frac{x+1}{y+1} > \frac{x}{y}$?

1. $x > 1$
2. $x < y$

Algebra Inequalities

Since y is positive, multiplying both sides of the inequality $\frac{x+1}{y+1} > \frac{x}{y}$ by $y(y+1)$ gives $y(x+1) > x(y+1)$ or $xy + y > xy + x$, which is equivalent to $y > x$. So, determining whether the inequality $\frac{x+1}{y+1} > \frac{x}{y}$ is true is equivalent to determining whether the inequality $y > x$ is true.

1. This indicates that $x > 1$. If, for example, $x = 3$ and $y = 4$, then $y > x$ is true. However, if $x = 4$ and $y = 3$, then $y > x$ is not true; NOT sufficient.
2. This indicates that$x < y$, so $y > x$ is true; SUFFICIENT.

The correct answer is B; statement 2 alone is sufficient.

•

If a school district paid a total of \$35 per desk for x desks and a total of \$30 per table for y tables, what was the total amount that the district paid for these desks and tables?

1. The total amount the district paid for the y tables was \$900.
2. $x = 90$, and the total amount the district paid for the x desks was 3.5 times the total amount the district paid for the y tables.

Algebra Simultaneous Equations

Determine the value of $35x + 30y$.

1. Given that $30y = 900$, it follows that $y = 30$. However, it is not possible to determine the value of $35x + 30y = 35x + 900$ because more than one possible value of x is possible; NOT sufficient.
2. Given that $x = 90$ and $35x = 3.5(30y)$, it follows that $(35)(90) = (3.5)(30)y$, or $y = \frac{35}{3.5} \times \frac{90}{30} = 10 \times 3 = 30$, and therefore $35x + 30y = 35(90) + 30(30)$; SUFFICIENT.

The correct answer is B; statement 2 alone is sufficient.

•

Three children inherited a total of X dollars. If the oldest child inherited \$7,000 more than the youngest child, and the youngest child inherited \$9,000 less than the middle child, what is the value of X?

1. The middle child inherited \$27,000.
2. The youngest child and the middle child together inherited a total of \$45,000.

Algebra Simultaneous Equations

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Algebra Simultaneous Equations

Let A, B, and C be the amounts inherited, respectively and in dollars, of the oldest, middle, and youngest child. It is given that $A = 7{,}000 + C$ and $C = B - 9{,}000$. Then $A = 7{,}000 + (B - 9{,}000) = B - 2{,}000$. Determine the value of $A + B + C$, or equivalently, determine the value of $(B - 2{,}000) + B + (B - 9{,}000) = 3B - 11{,}000$.

1. Given that $B = 27{,}000$, it follows that $3B - 11{,}000 = 70{,}000$; SUFFICIENT.
2. Given that $B + C = 45{,}000$, then, using $C = B - 9{,}000$, it follows that $B + (B - 9{,}000) = 45{,}000$, which can be solved to obtain $B = 27{,}000$. Therefore, $3B - 11{,}000 = 70{,}000$; SUFFICIENT.

The correct answer is D; each statement alone is sufficient.

•

If a building has 6,000 square meters of floor space, how many offices are in the building?

1. Exactly $\frac{1}{4}$ of the floor space is not used for offices.
2. There are exactly 20 executive offices, and each of these occupies 3 times as much floor space as the average for all of the remaining offices.

Algebra Simultaneous Equations

1. This says that exactly $\frac{3}{4}(6{,}000) = 4{,}500$ square meters of floor space are used for offices, since exactly $\frac{1}{4}(6{,}000) = 1{,}500$ square meters of the floor space are not used for offices. No information is given about the configuration of the 4,500 square meters used for offices. Therefore, the number of offices in the building cannot be determined; NOT sufficient.
2. While this says there are exactly 20 executive offices and each occupies 3 times the average space of the other offices, no information is given about how much of the 6,000 square meters of floor space in the building is used for offices nor how the offices are configured. Therefore, the number of offices in the building cannot be determined; NOT sufficient.

Taking (1) and (2) together, 4,500 square meters of space are used for offices which include 20 executive offices, and each executive office occupies 3 times as much floor space as the average of all of the remaining offices. However, no information is given about the actual size of an executive office or about the actual average size of the remaining offices. Therefore, the number of offices in the building cannot be determined. The examples in the table verify this:

Non-executive		Executive			
Number	Size (m^2)	Number	Size (m^2)	Total number of offices	Total space used for offices
30	50	20	150	30 + 20 = 50	30(50) + 20(150) = 4,500
15	60	20	180	15 + 20 = 35	15(60) + 20(180) = 4,500

The correct answer is E; both statements together are still not sufficient.

•

For which type of investment, J or K, is the annual rate of return greater?

1. Type J returns \$115 per \$1,000 invested for any one-year period, and Type K returns \$300 per \$2,500 invested for any one-year period.
2. The annual rate of return for an investment of Type K is 12%.

Arithmetic Percents

Compare the annual rates of return for the investment types J and K.

1. For Type J, the annual rate of return is \$115 per \$1,000 for any one-year period, which can be converted to a percent. For Type K, the annual rate of return is \$300 per \$2,500 for any one-year period, which can also be converted to a percent. These two percents can be compared to determine which is greater; SUFFICIENT.
2. Type K has an annual rate of return of 12%, but no information is given about the annual rate of return for Type J; NOT sufficient.

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J; NOT sufficient.

The correct answer is A; statement 1 alone is sufficient.

•

If Car X followed Car Y across a certain bridge that is $\frac{1}{2}$ mile long, how many seconds did it take Car X to travel across the bridge?

1. Car X drove onto the bridge exactly 3 seconds after Car Y drove onto the bridge and drove off the bridge exactly 2 seconds after Car Y drove off the bridge.
2. Car Y traveled across the bridge at a constant speed of 30 miles per hour.

Arithmetic Rate Problem

Find the number of seconds that it took Car X to cross the $\frac{1}{2}$-mile bridge.

1. If Car X drove onto the bridge 3 seconds after Car Y and drove off the bridge 2 seconds after Car Y, then Car X took 1 second less to cross the bridge than Car Y. Since there is no information on how long Car Y took to cross the bridge, there is no way to determine how long Car X took to cross the bridge; NOT sufficient.
2. If the speed of Car Y was 30 miles per hour, it took Car Y $\frac{1}{60}$ hour = 1 minute = 60 seconds to cross the bridge. However, there is no information on how long Car X took to cross the bridge; NOT sufficient.

Taking (1) and (2) together, Car X took 1 second less than Car Y to cross the bridge, and Car Y took 60 seconds to cross the bridge, so Car X took 60 − 1 = 59 seconds to cross the bridge.

The correct answer is C; both statements together are sufficient.

•

Will Candidate A win the most votes of any of the competing candidates in the upcoming election?

1. An opinion poll, Poll P, indicates that 60% of respondents prefer Candidate A over other candidates competing in the upcoming election.
2. An election-forecasting opinion poll is a method to estimate election outcomes based on a sample that is representative of the population of all of those legally entitled to vote.

Inference

The information provided is capable of providing a prediction of the election outcome based only on probabilistic data, as predictions usually are. So none of the information in the statements, taken either alone or together, can entail a yes or no answer to the question posed.

1. Of those who responded to Poll P, a subset of a sample of those polled, 60% preferred Candidate A over the other candidates. Was the sample of those eligible to vote truly representative? No answer is provided to this question—important for determining whether the poll is a reliable tool for making a prediction about the election outcome. Therefore, the information in the statement is insufficient by itself to determine a yes or no answer to the question posed; NOT sufficient.
2. An election-forecasting opinion poll provides data to **estimate** an election outcome. It can do so reliably only if it is based on a random sample representative of the whole population of eligible voters. The information in (2) is insufficient to answer the question posed about Candidate A; for example, it provides no data about the attitudes of voters toward Candidate A. Therefore, it cannot by itself entail an answer to the question posed about Candidate A's performance in the upcoming election; NOT sufficient.

(1) lacks the information to show that the poll mentioned in (1) conforms to the purpose and method of the type of election-forecasting poll that (2) describes. Therefore, (1) and (2), even combined, are insufficient to entail that Candidate A will win the upcoming election or will lose it.

The correct answer is E; both statements together are still not sufficient.

•

The government of Nation X makes policy decisions intended to advance the public interest, such as policy decisions aiming to promote employment, economic flourishing, and environmental protection. A policy decision can involve a

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significant trade-off, i.e., it can entail foreseeably giving up part or all of one valued outcome in the pursuit of another valued outcome. Does a policy decision by Nation X to permit extensive clearcutting of forests provide a valued outcome at the expense of another valued outcome?

1. Extensive clearcutting of forests is permitted in Nation X because it provides immediate economic gains by enabling businesses such as farming and lumbering to flourish and provide employment.
2. Extensive clearcutting of forests is permitted in Nation X even though it reduces forests' absorption of carbon dioxide, and this indirectly contributes to global warming, resulting in adverse climate changes that are certain to be very costly for Nation X to manage.

Evaluation

A policy decision involves a significant trade-off whenever the policy sacrifices one valued outcome in pursuit of another outcome that is also valued. Can the policy of permitting extensive clearcutting of forests entail a significant trade-off?

1. We are told that extensive clearcutting of forests in Nation X produces valued outcomes, but nothing is said or can be concluded about any potential trade-off. So (1) is not sufficient by itself to answer the question asked. This information, by itself, is not sufficient to indicate a trade-off; NOT sufficient.
2. This information indicates that permitting extensive clearcutting of forest sacrifices a valued outcome, but it does not enable us to conclude that permitting clearcutting of forests produces a valued outcome; NOT sufficient.

Even though (2) by itself is not sufficient to indicate a trade-off, the information in (1) and (2) combined is sufficient to indicate the trade-off involved in permitting extensive clearcutting of forests.

The correct answer is C; both statements together are sufficient.

•

Is Sarah's mathematics test score higher than the class average?

1. Sarah scored 90% on her mathematics test, answering 45 out of 50 questions correctly.
2. Sarah's high percentage score indicates that she performed exceptionally well, with only a few questions answered incorrectly.

Inference

The question posed is whether Sarah's mathematics test score is higher than the class average. We are given two pieces of information about Sarah's performance, but no information about how other people performed. Without the latter information, we cannot determine what the class average was.

1. Some information about Sarah's test score is given, but we are not given any information about the test scores of others in the class. So statement (1) is not sufficient alone; NOT sufficient.
2. Knowing that Sarah "performed exceptionally well" on her mathematics test does not allow us to infer that Sarah's performance was above the class average, even if Sarah's performance was exceptional for her. Statement (2) is not sufficient alone; NOT sufficient.

Taking (1) and (2) together does not give us enough information to answer the question asked. (1) does not help to remove the insufficiency described in (2).

The correct answer is E; both statements together are still not sufficient.

•

On Day X, was the forecast temperature 36°C at noon?

1. On Day X, the forecast temperature was 30°C at 9:00 a.m. and 40°C at 2:00 p.m.
2. On Day X, every half hour, the forecast temperature increased by 1°C.

Inference

1. This information provides that the forecast temperature was 30°C at 9:00 a.m., but nothing about the forecast temperatures after 9:00 a.m. except that the forecast temperature was 40°C at 2:00 p.m. By itself, it provides no information about the forecast temperature at noon. Therefore, statement (1) is not sufficient alone to provide a definitive answer to the question posed; NOT sufficient.
2. [illegible]

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2. The information that the forecast temperature increased by 1°C on each half hour would enable us to calculate the forecast temperature at each half hour after some earlier given time. But statement (2) gives us no forecast temperature for an earlier time in the 24-hour period. If statement (2) informed us, for example, that the temperature at 9:00 a.m. was 30°C, we could infer that it was 31°C at 9:30 a.m. This means that statement (2) is not sufficient by itself to answer definitively the question posed; NOT sufficient.

If we combine the information in statements (1) and (2), both of which refer to forecast temperatures on Day X, we can infer the forecast temperature at noon on Day X; since it increased by 2°C each hour, it was therefore 36°C at noon.

The correct answer is C; both statements together are sufficient.

•

Benjamin is a salesman working for commission at a shoe store. For the first $1,000 worth of sales in a given month, he earns a commission of 2%. For any sales he makes after the first $1,000 in that month, he earns a commission of 4%. What is the total dollar value of sales made by Benjamin in the month of March?

1. Benjamin earned a commission of $120 in March.
2. Benjamin doubled the dollar value of his sales in March compared to his sales in February.

Applied Problems

1. From the commission amount of $120 for March sales, we can infer the dollar value of those sales. He earned (2%)($1,000) in commission for the first $1,000 in sales. The remaining $100 in commission represented 4% of Benjamin's additional March sales, which therefore were (100 ÷ 0.04) dollars, i.e., $2,500. So total March sales were $(1,000 + 2,500), i.e., $3,500. The information provided is sufficient alone; SUFFICIENT.
2. The dollar value of Benjamin's March sales was double the value of his February sales. But this information is not sufficient alone to answer the question posed; NOT sufficient.

The correct answer is A; statement 1 alone is sufficient.

•

What is the length of Train A if it crosses Bridge X in 30 seconds from the moment it enters the bridge to the moment its final carriage exits the bridge?

1. Train B, which is 200 meters long, traveling at a constant speed of 20 meters per second, crosses Bridge X in 40 seconds.
2. The end of Train A, which is traveling at a constant speed, passes a pole in 10 seconds from the moment the train meets the pole.

Applied Problems

Data is presented about two trains, Train A and Train B, crossing Bridge X. Notice that a train has crossed a bridge when its total length has exited the bridge, so the crossing distance is the length of the bridge plus the length of the train. The question posed is what the length of Train A is.

1. We are given information about the constant speed of Train B, the length of Train B, its speed crossing the bridge, and how long that takes. That information about Train B is sufficient to establish the length of the bridge, which will be the total crossing distance minus the length of the train. But knowing this is not sufficient to infer the length of Train A, since we are not told the constant speed of Train A. So (1) is not sufficient by itself to enable us to infer a definite answer to the question posed; NOT sufficient.
2. We are not given the constant speed of Train A nor its length. (2) does not allow us to infer either of those magnitudes. If we knew its speed that enabled it to pass the pole in 10 seconds, then we could infer the train's length. So (2) is not sufficient alone; NOT sufficient.

Even combining the information in (1) and (2) will not be sufficient to allow the length of Train A to be inferred.

The correct answer is E; both statements together are still not sufficient.

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Solomon leases space for his barber shop, and his landlord has recently increased the rent by 40%. Solomon wants to increase his monthly revenue to pay the increased rent without affecting his net income. By what percentage must he increase his revenue?

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1. 20% of his previous monthly revenue was used to pay the rent before the landlord increased it.
2. The new rent is $50 more than the previous rent.

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Note that we are asked about a percentage, not a monetary amount.

1. We have enough information to calculate the percentage increase needed in Solomon's revenue. We have information from the stimulus about the percentage rent increase (40%) and from (1) about what percentage of revenue (20%) went to rent before the increase. To absorb that rent increase, Solomon's revenue will have to increase by 40% of that 20%, which is 0.4 multiplied by 0.2, i.e., 0.08, which is 8%. So the information in the stimulus together with that in (1) has enabled us to provide a definitive answer to the question asked. Therefore, (1) is sufficient alone; SUFFICIENT.
2. The information provided is that Solomon's rent increase is $50. Combining this information with the information in the stimulus allows us to calculate Solomon's old rent, R1, and his new rent, R2. If $50 = 0.4(R1), then R1 was $50/0.4, i.e., $125, and the new rent R2 is $175, i.e., $50 more than R1. But we lack enough information from the stimulus plus (2) to infer what percentage increase in revenue is needed to absorb the rent increase. We don't have enough information to calculate last year's revenue. So (2) is not sufficient alone; NOT sufficient.

The correct answer is A; statement 1 alone is sufficient.

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What is the ratio of the amount, by weight, of Metal A to Metal B in Alloy K?

1. The ratio of Metal A to Metal B in Alloy K becomes 5:6 when 40 pounds of Metal A is added to the alloy.
2. The ratio of Metal A to Metal B in Alloy K becomes 7:6 when 60 pounds of Metal A is added to the alloy.

Applied Problems

We are told how adding two different quantities of Metal A to Alloy K of Metals A and B changes the ratio between the two metals. We are asked to determine what the ratio between Metals A and B was before more of Metal A was added. Let A_1 be the quantity of metal.

1. Let A_1 be the original quantity in pounds of Metal A in the alloy before any further additions of Metal A. Let B_1 be the original quantity in pounds of Metal B in the alloy. Then, based on the information in (1), we know that $(A_1 + 40) : B_1 = 5 : 6$. Alternatively, this could be written as an equality of two fractional amounts: $(A_1 + 40)/B_1 = 5/6$. It follows from this that $5(B_1) = 6(A_1 + 40)$. This information is not sufficient alone to determine a unique value for the original ratio of Metal A to Metal B in Alloy K; NOT sufficient.
2. A similar approach to the information in (1) is applicable to the information in (2). This allows us to immediately conclude that the information in (2), like the information in (1), is also not sufficient alone to uniquely determine the original ratio of Metal A to Metal B in Alloy K; NOT sufficient.

The information gleaned from (2), as in (1), will give us an equation, derived in the same way as in (1). Then we have two equations, each with two unknowns, A_1 and B_1. Solving for A_1 and B_1 will allow us to provide a definitive answer to the question posed about the ratio of Metal A to Metal B in the original Alloy K before any further additions of Metal A. The answer you should get is the following: $A_1 : B_1 = 1 : 6$.

The correct answer is C; both statements together are sufficient.

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Train A started from Station X toward Station Y traveling at a constant speed, and on the same day, on a parallel track, Train B started from Station Y toward Station X at a constant speed. At what time did Train A reach Station Y?

1. The two trains met at 3:00 p.m. The distance traveled by Train A was twice the distance traveled by Train B by the time they met.
2. The ratio between the speeds of Trains A and B is 3:5. Train B started from Station Y at 6:00 a.m.

Applied Problems

We are asked to determine Train A's time of arrival at Station Y.

1. This provides us with the ratio between the distance traveled by Train A to the distance traveled by Train B when the trains meet at 3:00 p.m. But we lack enough information about the speeds of the trains, neither the specific

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the trains meet at 3:00 p.m. But we lack enough information about the speeds of the trains, neither the specific speeds nor the ratio between the speeds. Therefore, (1) is not sufficient alone to provide a definitive answer to the question posed; NOT sufficient.

2. This provides us with the ratio between the speeds of Trains A and B, and also tells us that Train B left Station Y at 6:00 a.m. But it tells us nothing about the distances that Trains A and B have traveled when they met. Therefore, (2) is not sufficient alone to provide a definitive answer to the question asked; NOT sufficient.

If the information in (1) and (2) is combined, it will be sufficient to determine the time at which Train A will reach Station Y. We have, in total, information concerning speeds, distances, and times. The distance Train A must travel from 3:00 p.m. is the distance Train B has already traveled. Train B covered that distance in 9 hours. The ratio between Train A's speed and Train B's speed is 3:5. So Train A will take 5/3 times 9 hours, i.e. 15 hours after 3:00 p.m., to arrive at Station Y. Therefore, the total information in (1) and (2) is sufficient to provide a definitive answer to the question, but neither (1) nor (2) is sufficient alone.

The correct answer is C; both statements together are sufficient.

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